A

Mini Project Report

On

AP Higher Education Intelligence using Python

Submitted to DR.K.V.SUBBA REDDY INSTITUTE OF TECHNOLOGY

In Partial Fulfillment of the requirements for the Award of Degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING(AIML/AI)

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UGC AUTONOMOUS

(Approved by AICTE, NEW DELHI, Affiliated to JNTUA, Anantapur)

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CERTIFICATE

This is to certify that the project entitled **"KVSR_Dashboard using Python"** is a bonafide work carried out by

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in partial fulfillment of the requirement for the award of the degree of **BACHELOR OF TECHNOLOGY** in **CSE(AI/AIML)** from Dr.K.V.Subba Reddy Institute of Technology, Kurnool, under our guidance and supervision.

The results presented in this Mini project have been verified and are found to be satisfactory. The results embodied in this Mini project have not been submitted to any other university for the award of any other degree or diploma.

Guide **Head of the Department**

DECLARATION

This is to certify that the work reported in the present Mini project entitled **"KVSR_Dashboard using Python"** is a record of bonafide work done by us in the Department of CSE(AI/AIML), Dr.K.V.Subba Reddy Institute of Technology, Kurnool. The reports are based on the project work done entirely by us and not copied from any other source. We submit our project for further development by any interested students who share similar interests to improve the project in the future.

The results embodied in this Mini project report have not been submitted to any other University or Institute for the award of any degree or diploma to the best of our knowledge and belief.

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Source distinction: the supplied *storytelling-dashboard(4).pdf* documents a sales storytelling project. Its question-pattern-action storytelling method is adapted here to the separate AP education dashboard documented by the ZIP. The screenshots supplied with this request belong to the AP dashboard, not to the sales project.

Scope of verification: Features below are supported by the ZIP README and Python source, and the four images show the supplied browser views. The screenshots are not evidence of official data or full acceptance testing.

01 Abstract

AP Higher Education Intelligence is a Streamlit-based interactive analytical web application designed to explore illustrative Degree and B.Tech college profiles across Andhra Pradesh. It converts a structured CSV/Excel dataset into KPI summaries, grouped bar graphs, gender donut charts, rankings, heatmaps, area charts and treemaps, allowing users to tell a coherent story about changes across academic years and selected categories.

The uploaded ZIP provides an illustrative directory of 78 colleges, 477 college-course links and 17,780 synthetic enrollment rows covering academic-year starts 2020 through 2026. These records are a repeatable teaching dataset and must never be interpreted as historical enrolment figures. Users can switch between executive, Degree and B.Tech analyses and examine filters for district, institution, course/branch, class year and section.

Keywords: Andhra Pradesh; higher education; Streamlit; Pandas; Plotly; data storytelling; Degree; B.Tech; synthetic data.

02 Introduction

2.1 Background and motivation

College information can be distributed across directories, spreadsheets and course lists. Comparisons become cumbersome when a user needs to isolate years, disciplines, institutions, gender composition and class sections. A browser-based dashboard combines these dimensions within one navigable interface.

2.2 Problem statement

Design a maintainable educational analytics prototype in which users can explore institution and program examples and understand distributions, changes over time and differences between Degree and B.Tech cohorts through interactive visual storytelling.

2.3 Objectives

Provide a statewide executive overview; separate Degree and B.Tech dashboards; expose male and female totals; compare academic years 2020-2026; display year/class and section data; offer searchable institution context and downloadable source/filtered files; preserve explicit synthetic-data labeling.

2.4 Scope

The project is an analytics and learning tool, not an admissions portal, an authenticated student management system, or a verified list of every college in Andhra Pradesh. The directory is partial and course associations are illustrative.

03 Existing System / Literature Survey

Conventional tools include spreadsheets for tabular investigation, static PDF reports for sharing findings, and enterprise BI portals for repeatable interactive analysis. Streamlit with Pandas and Plotly offers a lightweight Python alternative suitable for classroom prototyping. This comparison is conceptual; no performance benchmark was supplied.

Method	Relevance and limitations
CSV / Excel	Portable and editable, but multi-view chart updates require manual setup.
Static PDF	Effective for presentation, but filters and chart hover are unavailable.

Method	Relevance and limitations
Streamlit + Plotly	Unified widgets, calculations and charts; still needs trustworthy source data and deployment.

04 Technologies and Tools Used

Technology	Role
Python	Application logic, data generation and project utilities.
Streamlit	Web layout, sidebar controls, page selection, data tables, download widgets.
Pandas	CSV access, filtering, grouping, aggregation and reshaping.
Plotly Express / Graph Objects	Interactive bar, pie/donut, area, heatmap and treemap charts.
CSV and XLSX	Separated datasets and multi-sheet workbook for inspection/export.

05 System Requirements

5.1 Hardware

A modern Windows laptop with a recent supported Python installation, approximately 4 GB RAM or more, and browser access is suitable for an academic demonstration. This is a practical recommendation rather than a performance-tested minimum.

5.2 Python dependencies

requirements.txt specifies streamlit >=1.36,<2; pandas >=2.0,<3; and plotly >=5.20,<7. Use a Python version compatible with the installed dependency releases.

5.3 Windows installation

1. Extract AP_Education_Streamlit_Project ZIP.

2. Open Terminal in the AP_Higher_Education_Intelligence folder.

```
py -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python -m streamlit run app.py
```

Open http://localhost:8501 in the local browser.

If PowerShell blocks environment activation, run the environment’s python.exe directly for installation and launch, as detailed in README.md. Localhost works only on the computer running Streamlit unless network access is configured.

06 Dataset Description

6.1 Dataset origin and coverage

The package contains synthetic demonstration records created by generate_dataset.py with a fixed seed, a partial illustrative list of college names and course offerings, and a multi-sheet Excel workbook. The academic-year starts run from 2020 to 2026 (2020-21 through 2026-27). The package documentation describes 28 current 2026 AP district labels, including their retrospective use for previous years; this is not a historical district-boundary model.

Data item	Package status
78 college names	Illustrative, partial directory; not an official verified college census.
477 college-course links	Illustrative assignments, not a verified list of offered programs.
17,780 enrollment rows	100% fictional class, year, section and male/female statistics.
2020-2026	Seven academic-year starting labels.
Class/section	Study years 1-4 and sections A, B and C; degree fourth-year demo begins in 2024.

6.2 CSV and Excel inventory

Core files include colleges.csv, programs.csv, enrollments_2020_2026.csv, districts.csv and data_dictionary.csv. Separate BTech_ and Degree_ CSVs provide institution, course and enrollment partitions. Year/gender aggregate CSVs summarize college, branch and district totals. AP_College_Dashboard_Dataset_2020_2026.xlsx provides multiple workbook sheets.

6.3 Data governance

College names, categories, year labels and gender counts must be clearly distinguished from validated facts. Replacing demonstration rows requires authorized and responsibly aggregated data, consistent IDs, schema checks and updated workbook exports.

07 Implementation and Architecture

7.1 File organization

File or folder	Purpose
app.py	Streamlit dashboard layout, filters, aggregation and visuals.
generate_dataset.py	Creates deterministic synthetic CSV demonstration data.
data/	Source CSVs, summaries and Excel workbook.
requirements.txt	Package installation constraints.
README.md	Launch steps, features and data limitations.

7.2 Processing flow

Dataset CSVs are loaded into Pandas DataFrames. Sidebar selections define a subset of years, college type, districts, institutions, programs, study years and sections. The application calculates aggregate totals and creates Plotly charts from the filtered subset, then displays them in Streamlit widgets and exposes selected downloads.

7.3 User interface modules

The source includes Executive Overview, B.Tech Analytics, Degree Analytics, College Directory, Class & Section Insights, Dataset Explorer and Methodology. Navigation is provided through a Streamlit sidebar, with a dark-themed visual style and interactive Plotly chart controls.

7.4 Operational constraints

CSV inputs must remain present under the expected relative data folder. Regenerating the demonstration CSVs does not automatically refresh the pre-existing Excel workbook, as the README warns. Source-level review cannot establish uptime, correctness with real-world records, or complete browser coverage.

08 Data Visualization and Storytelling

8.1 Storytelling framework

Following the supplied sales-storytelling PDF’s general method, each analytical view can be explained in four stages: frame a question, show filtered evidence, state a defensible observation, and suggest a verification or follow-up action. Observation must be separated from any hypothesis about why the pattern exists.

8.2 Executive statewide overview

Question: How is the selected demonstration enrollment population divided across time, gender, college type and location? **Evidence:** cumulative KPI cards, stacked academic-year gender bars, a male/female donut, ranked institutions and course charts, a class-year area graph, district comparisons and a year-by-district heatmap. **Interpretation:** charts describe the generated demonstration population only.

8.3 B.Tech analytical story

Question: Which engineering branches and class sections contribute to the filtered sample? **Evidence:** B.Tech-only KPI cards, year and gender comparisons, branches and college rankings, section-based charts. **Action:** identify patterns worth verifying against authorized institutional records.

8.4 Degree analytical story

Question: How do the illustrative Degree college programs differ across selected years? **Evidence:** Degree-only bar and donut charts, program rankings and class-group summaries. **Caution:** do not interpret associations as an official inventory of active courses.

8.5 Visualization types

Visual	Interpretive purpose
Stacked bar	Compare male/female figures by academic year.
Donut chart	Describe selected male/female composition.
Horizontal ranking bars	Compare illustrative college and course aggregates.
Area chart	View selected class-year composition across time.
Heatmap	See year/district concentration in the demo.
Treemap	Explore hierarchical college type, district and branch.
Grouped/stacked class bars	Compare section and gender compositions.

09 Results and Discussion

The provided browser screenshots show the Executive Overview, B.Tech Analytics and Degree Analytics views. They demonstrate the dashboard structure, sidebar filtering, high-contrast metric cards, year-wise stacked gender charts and gender donut views. Values differ when the active date range or category changes, as expected for interactive filtering.

Observed in supplied screens	Interpretation
Executive screenshot: 281,596 records, 78 colleges, 477 course links	One selected three-year 2020-2022 view combining Degree and B.Tech.
B.Tech screenshot: 161,599 records, 38 colleges, 239 course links	Demonstration-only B.Tech view for 2020-2022.
Degree screenshot: 119,997 records, 40 colleges, 238 course links	Demonstration-only Degree view for 2020-2022.
Degree screenshot: 163,785 records	A second Degree view extends the selected date range to 2020-2023.

The figures in these screenshots are filtered cumulative totals for demonstration rows, not counts of unique human students and not current Andhra Pradesh statistics. Comparisons across screenshots must use matched filters and date boundaries. The storyboard can prompt questions, but the project does not establish causal drivers for enrollment changes.

9.1 Validation evidence and caveats

Inspection confirms the source package contains the application file, dependency declarations, dataset generator and separate CSV/XLSX artifacts. The supplied screenshots confirm these pages were displayed in a localhost browser. No live database test, government-data verification, accessibility audit, or automated UI regression report was provided.

10 Advantages and Limitations

10.1 Advantages

Separate Degree/B.Tech perspectives; multiple complementary visual types; centralized filtering; downloadable CSV/Excel data; clear dark-themed interface; reproducible synthetic examples; meaningful narrative pathway from KPI to drill-down.

10.2 Limitations

Partial college directory; unverified course mappings; entirely synthetic enrollments; current-district labels applied to historical examples; no validated admissions feed; absence of an authenticated production governance layer; mobile-browser views require separate usability checks; CSV regeneration does not automatically update Excel.

11 Conclusion

The AP Higher Education Intelligence prototype demonstrates how Pandas processing, Plotly charting and Streamlit interactions can turn multi-dimensional educational demonstration data into an accessible analytical story. It supports comparative exploration by institution type, date, gender, district, branch, class and section. Findings remain instructional and cannot be used as official institutional or state policy evidence without replacing and validating the source data.

12 Future Enhancements

Prioritize official year-matched college/course directories; approved aggregated enrollment imports; consistent historical district geography; schema and integrity validation; scheduled synchronized CSV/XLSX production; protected administration for authorized datasets; responsive and accessibility testing; provenance labels on every chart; optional exportable narrative reports. These are proposed enhancements, not claims about existing functionality.

13 References

Primary supplied sources: **AP_Education_Streamlit_Project(3).zip** (app.py, generate_dataset.py, README.md, data directory), **KVSR_Smart_Campus_Clean_Borders_Correct_Page_Numbers(3).pdf** (academic structure/layout reference), **storytelling-dashboard(4).pdf** (question-evidence-action storytelling approach) and four November 2026 dashboard screenshots.

Technical reference documentation: https://docs.streamlit.io/ ; https://pandas.pydata.org/docs/ ; https://plotly.com/python/ ; https://docs.python.org/3/ . These URLs are learning references, not proof of official enrollment data.

14 Appendix

Appendix A Representative program workflow

- Read college, course and enrollment CSV files
- Apply sidebar year/type/district/college/branch/class/section filters
- Compute male, female and total aggregate columns
- Render Plotly bar, donut, heatmap, area and treemap views
- Expose CSV/XLSX downloads and methodology disclosures

Appendix B Dataset field and source notes

Core joining keys use college_id and program_id. The demo analytics combine academic_year, college_type, district, branch/course, study-year/class, section, male, female and totals. Confirm the supplied data_dictionary.csv before changing column definitions.

Appendix C Additional Dashboard Screenshots

Four original user-supplied screenshots are reproduced on the next four pages. Captions describe the displayed filtered examples. They do not establish independent data verification.

14 Appendix C - C.1 Executive Overview - Statewide Sample Dashboard

Displays the 2020-2022 statewide illustration, 78 sample colleges and gender-wise bar/donut analytics.



Figure C.1. User-supplied Streamlit browser capture. Counts are synthetic demonstration records.

The statewide view brings 78 sample colleges together with 281,596 demo enrollment records.

The bars show year-wise male and female totals, while the donut summarizes their overall split.

This overview helps identify patterns to explore further using the district and college filters.

14 Appendix C - C.2 B.Tech College Analytics

Shows B.Tech-only records, 38 sample institutions and selected 2020-2022 male/female figures.

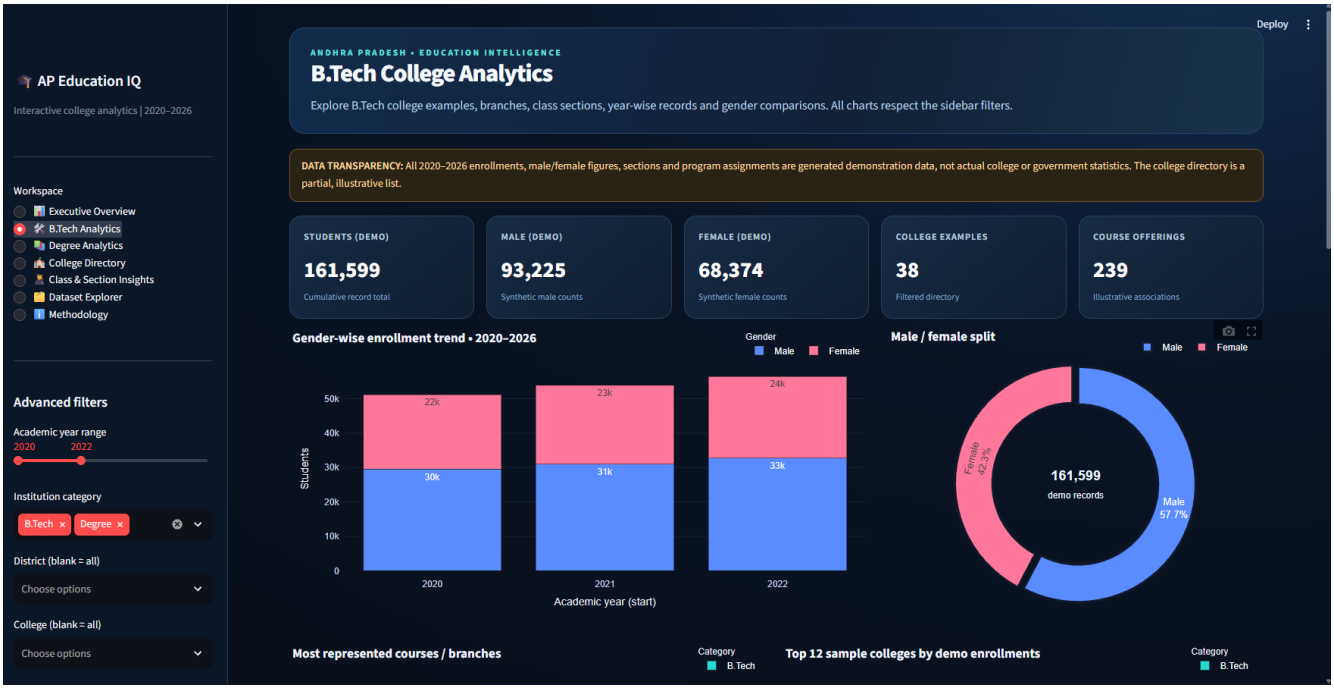


Figure C.2. User-supplied Streamlit browser capture. Counts are synthetic demonstration records.

The B.Tech dashboard focuses on 38 sample engineering colleges and 161,599 demo records.

Male and female counts are compared across 2020-2022 to reveal the selected sample composition.

These comparisons guide deeper branch-wise analysis, rather than claims about actual admissions.

14 Appendix C - C.3 Degree College Analytics - 2020 to 2022

Shows 40 Degree college examples, 238 course links and the three-year sample view.

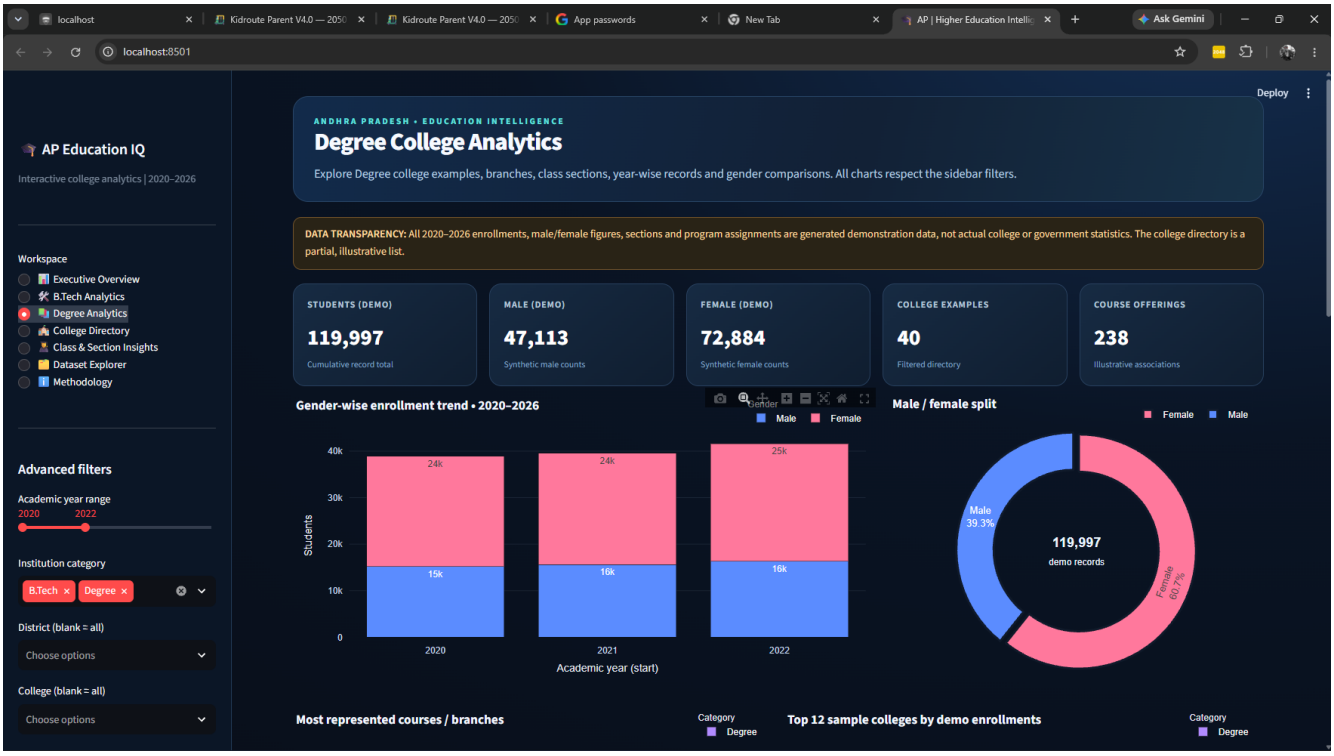


Figure C.3. User-supplied Streamlit browser capture. Counts are synthetic demonstration records.

The Degree view covers 40 sample colleges and 119,997 records across the 2020-2022 selection. Its stacked bars and donut show that female records form the larger share in this filtered view. Users can compare courses or colleges next to explore where the sample differences appear.

14 Appendix C - C.4 Degree College Analytics - Extended Year Range

Shows the Degree dashboard with the year filter extended to 2023, changing the demo totals.

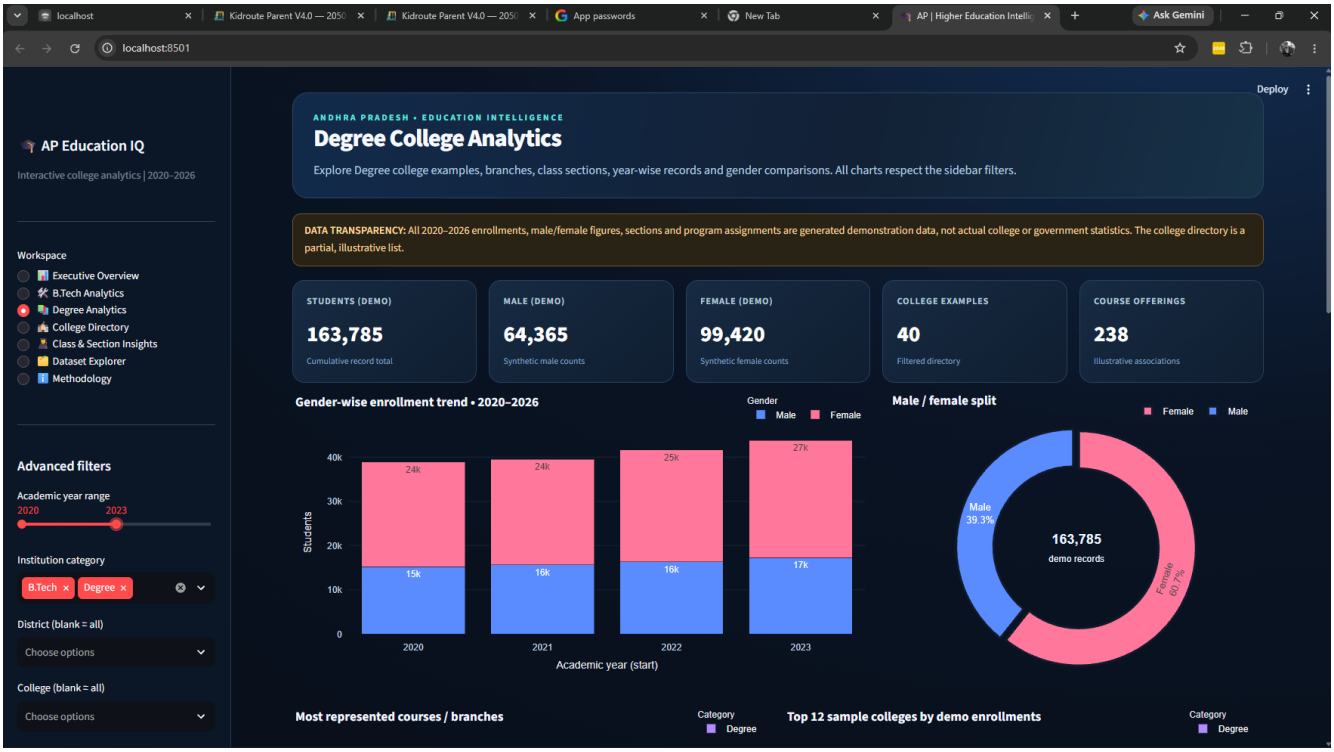


Figure C.4. User-supplied Streamlit browser capture. Counts are synthetic demonstration records.

Extending the Degree year range through 2023 increases the displayed total to 163,785 records.

The new year adds another bar and updates the gender donut to reflect the broader selection.

This demonstrates how time filters change the story; comparisons should use matching years.